Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression**	
Experiment No: 02	**Date:**	**Enrollment No: 92410133038**

Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression

IDE: Google Collab

Theory:

Linear regression is a method for determining the best linear relationship between two variables X and Y. If variables X and Y are uncorrelated, it is pointless embarking upon linear regression. However, if a reasonable degree of correlation exists between X and Y then linear regression may be a useful means to describe the relationship between the two variables.

Pre Lab Exercise:

A. Explain the meaning of linear regression

Linear Regression is a statical technical used to find the relationship between variables in ML context , linear regression finds the relationship between Features and Label.

B. Write three applications of linear regression

1. House Predictions based on Area
2. Predicting sales based onn advertisement cost
3. Predicting student marks based on their performances..etc..

C. Write three advantages of linear regression

1. It is simple and easy to understand.
2. It works well when there is a linear relationship between variables.
3. It is computationally efficient and fast to train.

D. Write three limitations of linear regression

1. It works only when the relationship between variables is linear.
2. It is sensitive to outliers.
3. It assumes constant variance and independence of errors.

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression**	
Experiment No: 02	**Date:**	**Enrollment No: 92410133038**

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Pre-process the data
5. Visualize the Data
6. Separate the feature and prediction value columns
7. Write the Hypothesis Function
8. Write the Cost Function
9. Write the Gradient Descent optimization algorithm
10. Apply the training over the dataset to minimize the loss
11. Find the best fit line to the given dataset
12. Observe the cost function vs iterations learning curve

Program (Code):

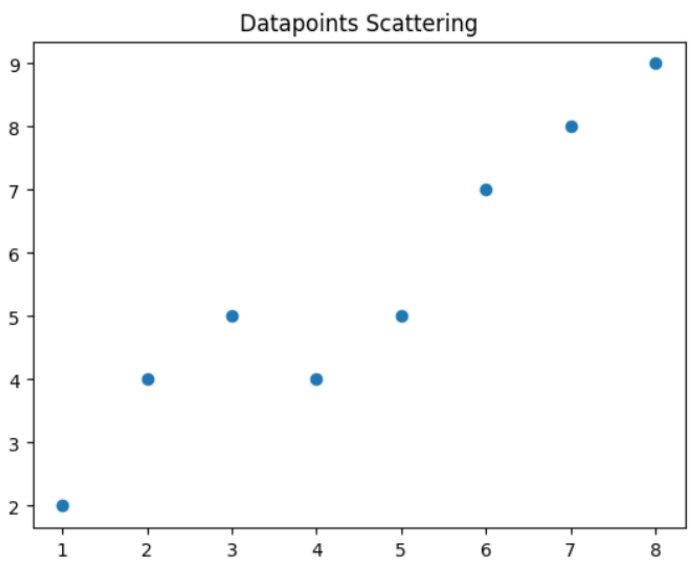
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

X = np.array([1,2,3,4,5,6,7,8])
Y = np.array([2,4,5,4,5,7,8,9])
m = 0
c = 0

learning_rate = 0.01
iterations = 1000
n = len(X)
cost_history = []

for i in range(iterations):
   Y_pred = m*X + c
   cost = (1/(2*n)) * np.sum((Y_pred - Y)**2)
   cost_history.append(cost)

      dm = (1/n) * np.sum((Y_pred - Y)*X)
   dc = (1/n) * np.sum(Y_pred - Y)
      m = m - learning_rate * dm
```

```
    c = c - learning_rate * dc
print("Final slope (m):", m)
print("Final intercept (c):", c)
```

After run:

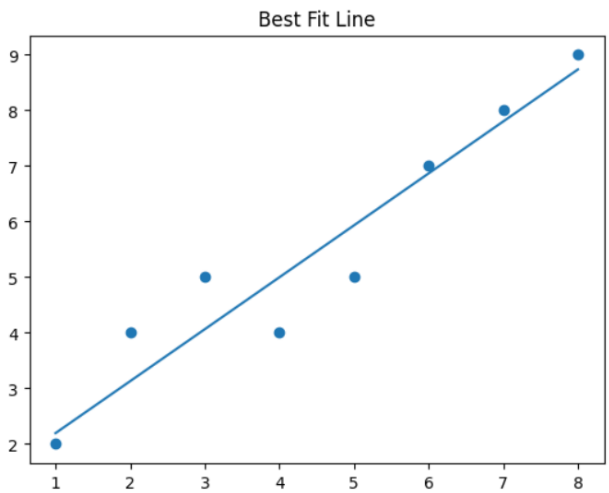
Final slope (m): 0.9343942778478435
Final intercept (c): 1.26196911916982

```
# Scatter plot without line
plt.scatter(X, Y)
plt.title("Datapoints Scattering")
plt.show()
```

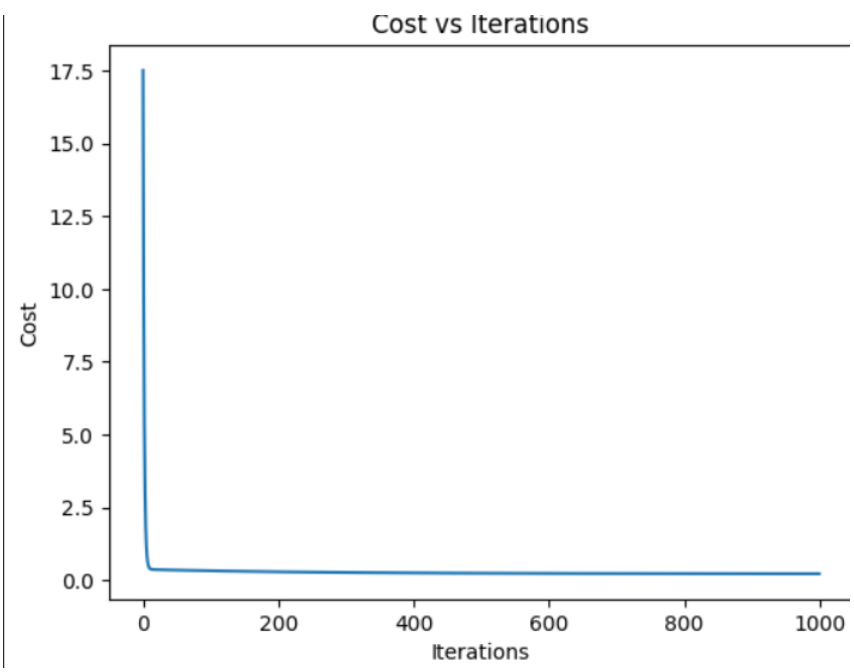



```
# Scatter plot with best fit line
plt.scatter(X, Y)
plt.plot(X, m*X + c)
plt.title("Best Fit Line")
plt.show()
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression**	
Experiment No: 02	**Date:**	**Enrollment No: 92410133038**




```
# Cost vs Iterations
plt.plot(cost_history)
plt.title("Cost vs Iterations")
plt.xlabel("Iterations")
plt.ylabel("Cost")
plt.show()
```



Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression**	
Experiment No: 02	**Date:**	**Enrollment No: 92410133038**

Program (Code & Outputs):-

Lab_02

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[1]
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

[2]
```python
X = np.array([1,2,3,4,5,6,7,8])
Y = np.array([2,4,5,4,5,7,8,9])
m = 0
c = 0
```

[3]
```python
learning_rate = 0.01
iterations = 1000
n = len(X)
cost_history = []
```

[5]
```python
for i in range(iterations):
    Y_pred = m*X + c
    cost = (1/(2*n)) * np.sum((Y_pred - Y)**2)
    cost_history.append(cost)
    dm = (1/n) * np.sum((Y_pred - Y)*X)
    dc = (1/n) * np.sum(Y_pred - Y)
    m = m - learning_rate * dm
    c = c - learning_rate * dc
```

```python
print("Final slope (m):", m)
print("Final intercept (c):", c)
```

```
Final slope (m): 0.9343942778478435
Final intercept (c): 1.26196911916982
```

```python
# Scatter plot without line
plt.scatter(X, Y)
plt.title("Datapoints Scattering")
plt.show()
```

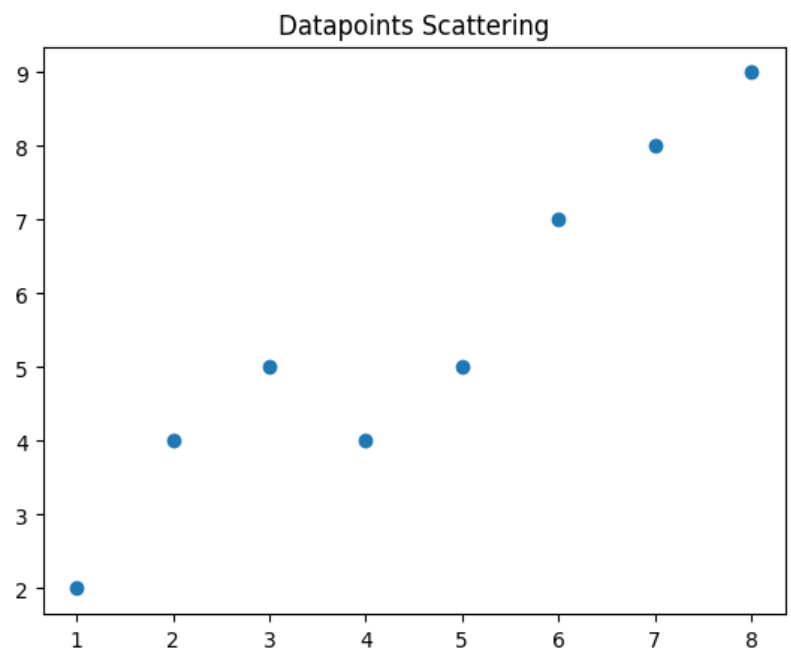
```python
# Scatter plot with best fit line
plt.scatter(X, Y)
plt.plot(X, m*X + c)
plt.title("Best Fit Line")
plt.show()
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression**	
Experiment No: 02	**Date:**	**Enrollment No: 92410133038**

```
# Cost vs Iterations
plt.plot(cost_history)
plt.title("Cost vs Iterations")
plt.xlabel("Iterations")
plt.ylabel("Cost")
plt.show()
```

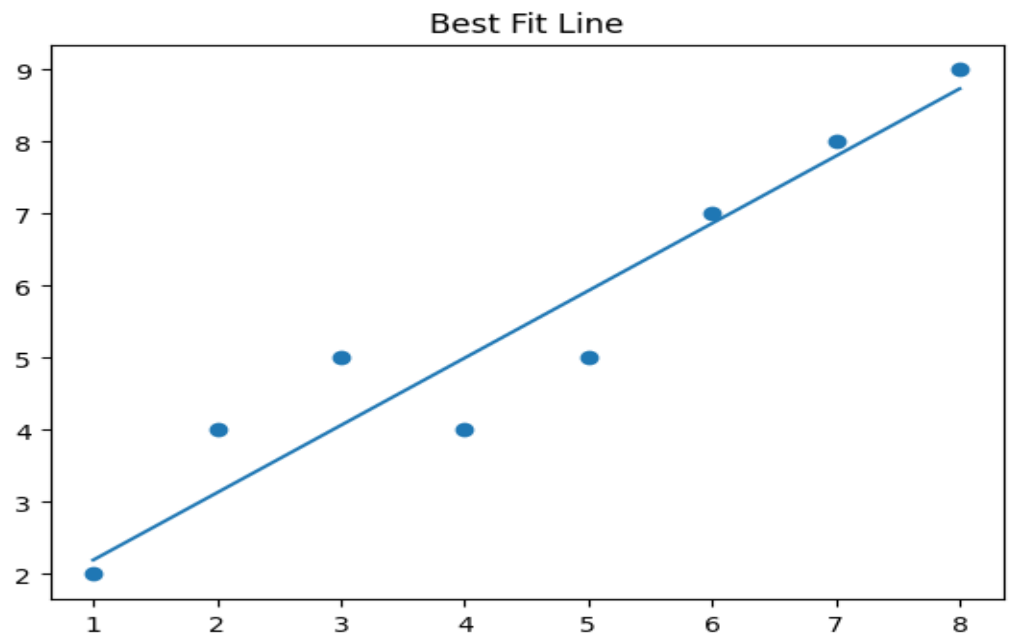
Results:

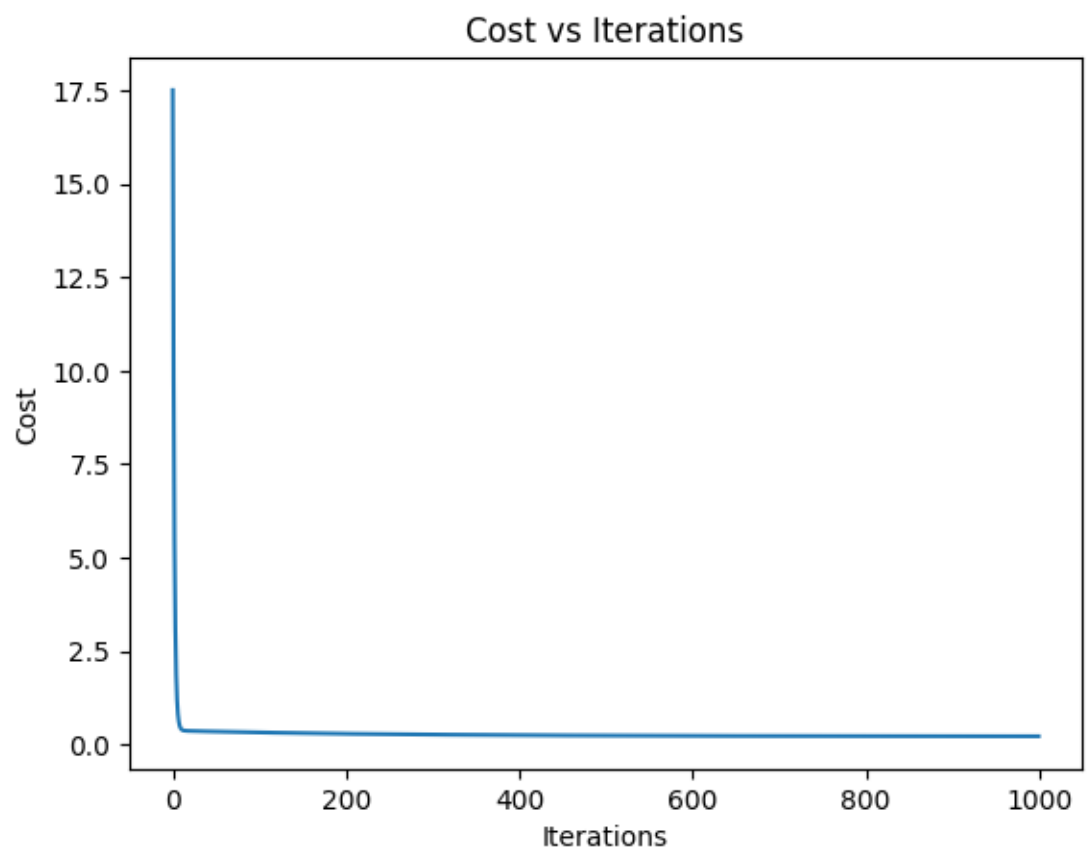
a. Datapoints scattering (without best fit line)



b. Scatter plot showing the Best fit line in the last iteration of training

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression**	
Experiment No: 02	**Date:**	**Enrollment No: 92410133038**





Observation and Result Analysis:

A. Nature of the dataset

The dataset contains single feature input values and corresponding output values. From the scatter plot, it can be observed that the data points show a roughly linear pattern. This indicates that linear regression is suitable for modeling the relationship between the variables

B. During Training Process

During the training process, gradient descent was applied to minimize the cost function. In each iteration, the values of m (slope) and c (intercept) were updated. The cost function value gradually decreased with each iteration. This shows that the model was learning and moving towards the global minimum._ The cost vs iterations graph shows a decreasing curve, which indicates proper convergence of the algorithm

C. After the training Process

After training, the model found the optimal values of m and c which define the best fit line. The final scatter plot shows the straight line passing close to most of the data points. The difference between predicted and actual values was minimized. This confirms that the model successfully learned the relationship between the feature and the target variable

<u>Post Lab Exercise:</u>

A. What are the major assumptions considered in linear regression

1. There should be a linear relationship between independent and dependent variables.
2. The errors should be independent.
3. The variance of errors should be constant (homoscedasticity).
4. The errors should be normally distributed (in many cases).

B. Why MSE is used instead of MAE for calculating the loss function

MSE (Mean Squared Error) is used because it penalizes larger errors more than smaller errors due to squaring. It is also mathematically convenient because it is differentiable, which makes gradient descent easier to apply. MAE is not as smooth as MSE, so optimization becomes slightly more difficult

C. How can the behaviour of outliers be understood while dealing with the unseen dataset

Outliers can be identified using visualization techniques like scatter plots or box plots. If some data points lie far away from the general pattern, they are considered outliers. Since linear regression is sensitive to outliers, they can significantly affect the slope of the line. Therefore, it is important to detect and handle them properly before training

D. Derive the Normal Equation for the Linear Regression.

For linear regression, the hypothesis is:

$$\mathbf{y = X\theta}$$

The cost function is:

$$\mathbf{J(\theta) = (1/2m)\,(X\theta - y)^2}$$

To minimize the cost function, we take the derivative with respect to θ and set it equal to zero.

After simplification, we get the Normal Equation:

$$\boldsymbol{\theta = (X^T X)^{-1} X^T y}$$

This equation directly gives the optimal values of parameters without using gradient descent.

Conclusion:

In this experiment, Linear Regression was implemented to find the best fit line for the given dataset. The relationship between the feature and target variable was analyzed using scatter plots. The model was trained using gradient descent to minimize the cost function and obtain optimal values of slope and intercept. After training, the best fit line was plotted which represented the linear relationship between the variables. This experiment helped in understanding how linear regression works and how it can be used for prediction tasks.